

8. A shell script is a file that contains
 (A) Set of statements (B) Scheduling algorithm
 (C) Interrupt routine (D) Sequence of operating system commands and shell commands
9. The hardware implementation which provides mutual exclusion is
 (A) Counting semaphore (B) Binary semaphore
 (C) Test and set lock (D) Scheduling algorithm
10. The reader-writer problem and its solutions, have been generalized to provide reader writer locks on some systems. The process wishing to modify the shared data must request the lock in _____.
 (A) Read mode (B) Write mode
 (C) Read and write mode (D) Exclusive mode
11. The circular wait condition can be prevented by _____.
 (A) Defining a linear ordering of resources type (B) Using thread
 (C) Using pipes (D) Thread and pipes
12. For non sharable resources like a printer, mutual exclusion _____.
 (A) Must exist (B) Must not exist
 (C) May exist (D) Partially exist
13. The last on the hierarchy scale of memory device is _____.
 (A) Main memory (B) Secondary memory
 (C) TLB (D) Flash drives
14. In contiguous memory allocation _____.
 (A) Each process is contained in a single contiguous section of memory (B) All process are contained in a single contiguous section of memory
 (C) The memory space is contiguous (D) The memory space is not contiguous
15. Internal fragmentation occurs when a _____.
 (A) Memory area remain unused because it is too large to be allocated (B) Memory area remain unused because it is too small to be allocated
 (C) More memory is allocated than requested by processor (D) Less memory is allocated than requested by processor
16. In virtual memory, swap space exist in _____.
 (A) Cache (B) Registers
 (C) RAM (D) Secondary storage
17. Which one of the following connects high-speed high-bandwidth device to memory subsystem and CPU?
 (A) Expansion bus (B) PCI bus
 (C) SCSI bus (D) External bus

18. After the completion of DMA transfer, the processor is notified by _____.
 (A) Acknowledgement signal (B) Interrupt signal
 (C) WMFC signal (D) Control signal

19. A file is a/an _____ data type.
 (A) Abstract (B) Primitive
 (C) Public (D) Private

20. The process of dividing a disk into sectors that the disk controller can read and write, before a disk can store data is known as _____.
 (A) Partitioning (B) Swap space creation
 (C) Low level formatting (D) Fragmentation

PART – B (5 × 4 = 20 Marks)

Answer ANY FIVE Questions

Marks BL CO PO

21. Define process and explain the process states with neat diagram. 4 2 1 1
22. Explain the priority scheduling algorithm with suitable example and write the pros and cons. 4 3 2 2
23. Explain Peterson's solution for achieving mutual exclusion. 4 3 3 2
24. What is paging? Discuss basic paging technique in detail. 4 3 4 2
25. Discuss the difference between internal and external fragmentation. 5 3 5 2
26. Describe, using a good example, the effects of changing the Round Robin scheduling algorithms time quantum from fixed small number to fixed large number. 4 3 2 2
27. What is virtual memory? How it is implemented? Explain. 4 3 4 1

PART – C (5 × 12 = 60 Marks)

Answer ALL Questions

Marks BL CO PO

28. a. Describe the features of the following types of operating systems structures.
 (i) Layered approach
 (ii) Micro kernel system structure
- (OR)
- b. Describe the fields in Process Control Block (PCB). What is context switch overhead? Explain with neat sketch. 12 3 1 2
29. a. Suppose the following processes arrive for execution at the times indicated. Each process will run the listed amount of time, in answering the questions use non-preemptive scheduling. 12 4 2 2